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Career Objective

B.Tech Computer Science and Engineering candidate at College of Engineering Trivandrum with a strong foundation in Artificial Intelligence, intelligent systems, and AI-driven application development. Passionate about web development and engineering scalable, user-focused solutions through modern technologies. Focused on software engineering practices, problem solving, and delivering efficient systems through coding, backend logic, and cross-functional technical projects.

Education

B.Tech in Computer Science and Engineering

2024 – Present

College of Engineering Trivandrum (CET), Kerala

Semester 1 SGPA: 8.87 Semester 2 SGPA: 9.03 Semester 3 SGPA: 8.56 **Current CGPA: 8.79 / 10**

Relevant Coursework: Artificial Intelligence, Data Structures and Algorithms, Machine Learning, Web Development, Linear Algebra, Probability and Statistics, Object-Oriented Programming, Computer Networks

Technical Skills

- **Programming Languages:** Python, C, C++, JavaScript, HTML, CSS, Java, SQL, Bash
- **AI/ML Tools and Platforms:** Google AI Studio, OpenAI API, TensorFlow, Hugging Face Hub, LiteLLM, ChromaDB, LightFM, UMAP, sentence-transformers, Google Colab
- **Frameworks and Platforms:** FastAPI, React, Vite, Node.js, LlamaIndex, RAG Pipelines, REST APIs
- **Core Areas:** Machine Learning, Explainable AI, Recommendation Systems, Multi-Agent Systems, Web Development, System Design
- **Developer Tools:** VS Code, Git, GitHub, LaTeX, SQLite, Firebase, API Testing
- **Problem Solving:** LeetCode, HackerRank, CodeChef

Experience

Research Intern, Explainable Recommendation using Embeddings

May 2026 – June 2026

College of Engineering Trivandrum (CET), Kerala

- Developed an explainable movie recommendation framework using the LightFM hybrid model, leveraging user-item interactions and feature-based embeddings under faculty supervision.
- Built an explanation module across 4 complementary signals (embedding similarity, shared genres, user-generated tags, and collaborative evidence from similar users), improving recommendation transparency by surfacing human-interpretable reasoning for each prediction.
- Executed exploratory data analysis across 3 experiment types (UMAP-based embedding visualization, arithmetic probing, and cluster inspection) to uncover semantic structure in latent space.
- Evaluated recommendation performance across 4 metrics (Precision at K, Recall at K, AUC, and MRR) and reviewed recent XAI literature to align implementation with current research.

Projects

Multi-Agent Research Intelligence System (Techashy Hackathon)

2026

- Architected a reflective multi-agent pipeline using DeepSeek R1, Groq Llama 3.3 70B, and Cerebras Llama 3.1 70B for structured multi-perspective research analysis.
- Structured a sequential agent architecture (Supervisor, Pros, Cons, Future) where each agent builds on prior outputs to improve reasoning consistency.
- Integrated ArXiv API with sentence-transformers and ChromaDB for persistent RAG; implemented LiteLLM-based multi-provider routing with failover for reliability.

Blind Server Architecture for Anonymous Learning Platform (TechThrive 2.0)

2026

- Created a Zero-Knowledge-inspired adaptive learning platform with complete client-side quiz evaluation, level computation, and profiling.
- Engineered a stateless FastAPI backend serving modular JSON content with no sessions, database, or user data storage.
- Applied Fernet encryption with timestamp-based token validation for secure content delivery without exposing user behavior.

Kashmiri Multilingual Audio Intelligence Agent (GNOSIS Hackathon, Incepta 2026)

2026

- Won 3rd Place by constructing an offline multilingual audio analysis system for Kashmiri, Hindi, and Urdu speech,

competing against 20+ teams.

- Engineered a multi-stage pipeline including noise reduction, Faster-Whisper ASR, MarianMT translation, spaCy NER, and rule-based risk scoring.
- Implemented keyword-based threat detection with contextual flagging and a deterministic 3-tier risk classification system.

AI Semiconductor Datasheet Chatbot (Via CampusConnect Hackathon)

2025

- Won 1st Place by building an AI-powered RAG chatbot enabling natural-language querying of semiconductor datasheets.
- Processed PDF datasheets into semantic embeddings using BGE models stored in Qdrant for efficient similarity-based retrieval.
- Integrated LLMs (TinyLlama, Qwen) with LlamaIndex to generate accurate, context-aware responses grounded in datasheet content.

Achievements

- **3rd Place, GNOSIS Hackathon (Incepta 2026):** Built the Kashmiri Multilingual Audio Intelligence Agent, competing against 20+ teams.
- **1st Place, Via CampusConnect Hackathon:** Delivered an AI-based RAG solution for semiconductor chip datasheet analysis.
- Maintained strong academic performance with CGPA 8.73/10 across 3 semesters, placing in the top tier of the cohort.
- Led cross-functional teams as Team Lead across 4 hackathons, delivering AI and software solutions under competitive time constraints.
- Resolved 100+ algorithmic problems on LeetCode, HackerRank, and CodeChef using Python and C, demonstrating strong command of data structures and algorithms across arrays, graphs, dynamic programming, and recursion.
- Shipped full-stack web and AI applications using React, Firebase, FastAPI, and Python, delivering REST APIs, backend systems, and integrated UI/UX workflows across multiple independent projects.

Activities and Leadership

- **Open Source Contributor, GSSoC, NSoC, SSoC:** Actively participated in open source programs under GirlScript Summer of Code (GSSoC), Nexus Spring of Code (NSoC), and Social Summer of Code (SSoC), collaborating with global maintainers on real-world codebases.
- **Web Team Member, AstroCET (Astronomy Club, CET):** Developed and maintained the club website, implementing UI fixes and content updates using React.
- **Content Subpanel Member, CS Association CET:** Produced technical content, supported event organization, and managed communication for the department's student association.
- **Technical Team Member, Glitch CET (Game Development Club):** Delivered front-end integration and interactive systems across 2+ collaborative projects, improving interface responsiveness and reducing interaction latency.
- **Technical Team Member, Build Club CET:** Engaged in collaborative software projects, technical discussions, and rapid prototyping, shipping 3+ prototypes across different problem domains.
- **Volunteer, Dotslash 2025 (CET):** Assisted in organizing and coordinating one of CET's flagship technical fest events, supporting smooth execution across sessions and activities.

Certifications and Workshops

- Solutions Challenge, Google Developer Group CET
- Web Development and AI with TensorFlow, Dotslash 2025 CET
- Ctrl+Create Game Development Challenge, Glitch CET
- Python, Blender, and MS Excel Workshop, SAEIndia CET

Soft Skills

- Analytical and Creative Problem Solving
- Agile Team Collaboration
- Curiosity-Driven Learning
- Project Management and Accountability

Interests

Artificial Intelligence, Web Development, Software Engineering, Machine Learning, Cybersecurity, AI-driven Security, Multi-Agent Systems, Explainable AI, Hackathons, Technology Trends, Reading, Photography